

# 1 Static Yao-Kivelson Model

## 1.1 Kitaev Model and Yao-Kivelson Model

Kitaev found a way to map the strongly-interacting spin model

$$H = - \sum_{\substack{\text{x-bonds} \\ \langle jk \rangle}} J_{jk}^x \sigma_j^x \sigma_k^x - \sum_{\substack{\text{y-bonds} \\ \langle jk \rangle}} J_{jk}^y \sigma_j^y \sigma_k^y - \sum_{\substack{\text{z-bonds} \\ \langle jk \rangle}} J_{jk}^z \sigma_j^z \sigma_k^z, \quad (1.1)$$

now known as the Kitaev Model, to a free Majorana-Fermion hopping model, and therefore managed to solve the model exactly [1]. Classically, 'the ' Kitaev model is defined on a honeycomb lattice, where each coupling type  $\alpha = x, y, z$  corresponds to one of the three bond orientations. However, his method directly generalizes to any tricoordinate lattice geometry [2], provided that each site has exactly one bond of each type  $\alpha = x, y, z$ .

One alternative to the honeycomb lattice is the star/triangle-honeycomb lattice. Instead of elementary plaquettes of size 6, it features two different types of plaquettes of size 3 and 12, respectively. When defined on this geometry, model 1.1 is known as the Yao-Kivelson Model [3]. Due to the odd length of the triangular plaquettes, this model features a groundstate that spontaneously breaks Time-Reversal Symmetry, and therefore describes a Chiral Spin Liquid.

## 1.2 Kitaev's Spin-Majorana Mapping

Kitaev introduced a representation of a spin- $\frac{1}{2}$ -degree of freedom (called 'spin' in the following) in terms of 4 fermionic Majorana modes. Consider the 4-dimensional Fockspace  $\mathcal{H}$  of two independent abstract fermions  $a_1, a_2$  satisfying the canonical anti-commutation relations

$$\left\{ a_k, a_l^\dagger \right\} = \delta_{kl}, \quad \left\{ a_k, a_l \right\} = 0. \quad (1.2)$$

They give rise to 4 real fermion modes

$$c_{2k-1} := c_k + c_k^\dagger \quad \text{and} \quad c_{2k} := \frac{1}{i} \left( a_k - a_k^\dagger \right), \quad \text{for} \quad k = 1, 2, \quad (1.3)$$

which are commonly called Majorana modes. They satisfy

$$c_i^\dagger = c_i \quad \text{and} \quad \{c_i, c_j\} = 2\delta_{ij}, \quad (1.4)$$

which qualifies them to be called "real fermions", in contrast to the usual complex ones.

The goal is now to use these to represent the spin algebra

$$\sigma^\alpha \sigma^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma} \sigma^\gamma, \quad \alpha, \beta, \gamma = x, y, z. \quad (1.5)$$

For that, rename the 4 Majorana modes from  $\{c_i\}_{i=1,\dots,4}$  to  $\{b^x, b^y, b^z, d\}$  and define

$$\tilde{\sigma}^\alpha := ib^\alpha d, \quad \alpha = x, y, z. \quad (1.6)$$

As  $(\tilde{\sigma}^\alpha)^2 = 1$  follows directly from eq. 1.4, eq. 1.5 reduces to the simpler, equivalent condition  $\tilde{\sigma}^x \tilde{\sigma}^y \tilde{\sigma}^z \stackrel{!}{=} i$ , which implies

$$1 \stackrel{!}{=} b^x b^y b^z d =: D, \quad (1.7)$$

where we have defined the hermitean operator  $D$ . Due to  $D^2 = 1$ ,  $\mathcal{H}$  splits into the eigenspaces  $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_{-1}$ , where the subscript denotes the respective eigenvalues of  $D$ . Consequently,  $\mathcal{H}_1$  is the space where eq. 1.7 is fulfilled, and is therefore called the 'physical' subspace. The projector into this subspace is given by

$$P = \frac{1 + D}{2}. \quad (1.8)$$

$D$  cannot be the identity on the whole Hilbert space  $\mathcal{H}$ , as it anti-commutes with each Majorana operator:

$$\{D, c_i\} = 0 \quad \text{for} \quad c_i = b^x, b^y, b^z, d. \quad (1.9)$$

It follows that each Majorana operator  $c_i$  maps 1-to-1 between  $\mathcal{H}_1$  and  $\mathcal{H}_{-1}$ , which shows that  $\dim(\mathcal{H}_1) = \dim(\mathcal{H}_{-1}) = 2$ . Equation 1.9 further implies that  $D$  commutes with operators consisting of an even number of Majorana operators. In particular,

$$[D, \tilde{\sigma}^\alpha] = 0, \quad \alpha = x, y, z, \quad (1.10)$$

which means  $D$  and one (but not all)  $\tilde{\sigma}^\alpha$ , say  $\tilde{\sigma}^z$ , can be diagonalized simultaneously, such that we arrive at an orthonormal basis  $|D = \pm 1, \tilde{\sigma}^z = \pm 1\rangle$  spanning  $\mathcal{H}$ :

$$\mathcal{H} = \text{span}\{\underbrace{|1, 1\rangle, |1, -1\rangle}_{\text{physical}}, |-1, 1\rangle, |-1, -1\rangle\}. \quad (1.11)$$

This motivates a unitary transformation from the usual spinor space  $\text{span}\{|\uparrow\rangle, |\downarrow\rangle\}$  to the physical subspace  $\mathcal{H}_1$  defined by

$$U|\uparrow\rangle := |1, 1\rangle, \quad \text{and} \quad U|\downarrow\rangle := |1, -1\rangle. \quad (1.12)$$

As the  $\tilde{\sigma}^\alpha$  preserve both subspaces respectively due to eq. 1.10, their restriction to  $\mathcal{H}_1$  finally defines a representation of the spin algebra sufficing

$$\sigma^z = U^\dagger \tilde{\sigma}^z U, \quad (1.13)$$

$$[D, H] = [P, H] = 0 \quad (1.14)$$

for any physical Hamiltonian  $H$  expressed in terms of the  $\sigma^\alpha$  from eq. 1.6. This is what allows for the following procedure to obtain eigenstates of a Hamiltonian:

1. Write  $H$  in terms of 4 Majorana modes per spin according to eq. 1.6.
2. Find the spectrum and eigenstates on the enlarged Hilbertspace.
3. Apply  $P$  to project into the physical subspace. The result will be an eigenstate of the physical Hamiltonian.

# Bibliography

- <sup>1</sup>A. Kitaev, “Anyons in an exactly solved model and beyond”, *Annals of Physics* **321**, 2–111 (2006).
- <sup>2</sup>T. Bauer and J. Reuther, “Compact localized fermions and ising anyons in a chiral spin liquid”, *Physical Review B* **113**, 10.1103/dgyy-rbcn (2026).
- <sup>3</sup>H. Yao and S. A. Kivelson, “Exact chiral spin liquid with non-abelian anyons”, *Physical Review Letters* **99**, 10.1103/physrevlett.99.247203 (2007).